



سوق المزاد - SouqAlmazad

Database Normalization Report

DataBase - Course Project

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Introduction

The SouqAlMazad database was designed as a realistic marketplace and auction platform that supports Buyers, Sellers, Admins, Products, Auctions, Orders, Payments, Shipments, Reviews, Wishlists, Notifications, and Reports. The schema was carefully normalized to reduce redundancy, improve consistency, preserve referential integrity, and support scalability.

First Normal Form (1NF)

A relation is in First Normal Form when every attribute contains atomic values and no repeating groups exist.

User Table

- Each user stores a single first_name, last_name, email, and phone value.

Product Images

- Instead of storing multiple image URLs inside the product table, a dedicated product_image table was created.

Wishlist

- Each wishlist item is stored as a separate row instead of comma-separated values.

Notifications

- Notifications are stored individually per user.

Non-Normalized	Normalized
product.images = 'img1,img2,img3'	product_image(image_id, product_id, image_url)

Second Normal Form (2NF)

A relation is in Second Normal Form when it satisfies 1NF and all non-key attributes depend entirely on the primary key.

Order_Item Table

- quantity, unit_price, and subtotal depend on the entire order_item_id.

Wishlist Table

- Each wishlist record depends on wishlist_id and represents one buyer-product relation.

Product_Image Table

- display_order and is_primary depend entirely on image_id.

The project uses AUTO_INCREMENT primary keys such as user_id, product_id, order_id, auction_id, and payment_id, which eliminates partial dependencies.

Third Normal Form (3NF)

A relation is in Third Normal Form when it satisfies 2NF and no transitive dependency exists between non-key attributes.

Category Normalization

- Category information is separated into the category table and referenced through category_id foreign keys.

ISA Specialization

- Buyer, Seller, and Admin data are separated into subtype tables instead of placing all fields in one large users table.

Order Information

- Buyer names and emails are not duplicated inside the order table. They are retrieved using JOIN operations.

ISA Specialization

The schema uses ISA specialization where User acts as the supertype while Buyer, Seller, and Admin act as subtype entities.

Subtype	Specialized Attributes	Purpose
Buyer	loyalty_points, date_of_birth, gender	Purchases and bidding
Seller	business_name, commission_rate	Store and product management
Admin	role, permissions	System administration and moderation

Controlled Denormalization

Some calculated values were intentionally cached for performance optimization. Examples include auction.current_highest, auction.winner_buyer_id, and store.rating_avg. Although these values could be dynamically calculated using aggregate queries, storing them directly improves dashboard and auction performance.

Referential Integrity and Constraints

- Primary Keys are used in all tables.
- Foreign Keys maintain relationships between entities.
- ENUM constraints enforce valid status values.
- CHECK constraints validate review ratings between 1 and 5.
- ON DELETE CASCADE is used for dependent entities.
- ON DELETE SET NULL is used for auction winners.

Transactions and ACID Properties

Transactions are heavily used in order creation, direct purchases, auction bidding, and auction closing. SELECT ... FOR UPDATE is used to prevent race conditions during bidding and stock updates.

ACID Property	Implementation Example
Atomicity	Orders and payments are created in one transaction.
Consistency	Foreign keys and validations maintain integrity.
Isolation	FOR UPDATE locking prevents concurrent conflicts.
Durability	Committed transactions remain permanently stored.

Conclusion

The SouqAlMazad schema was normalized up to Third Normal Form (3NF) while maintaining practical performance optimizations. The final database design supports marketplace operations, live auctions, order management, payments, shipments, notifications, moderation, and role-based access control in a consistent and scalable relational architecture.